Orientation, set-up, and your first app

Environment, tooling and a first app on a device

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Lab objectives

Install a working toolchain

Flutter and Dart on your laptop, with flutter doctor reporting no blocking issues.

Run an app on a device

An emulator or simulator that boots, with the default Flutter app running on it.

Time
2 hours

Present
To the lab instructor, at the end of this lab

The cumulative lab app

1. One Flutter project per student. Every lab continues it.
2. At the end of each lab, present what you built to the lab instructor.
3. Install the tools before starting.

Seven labs build one app that grows from empty to networked and authenticated. Lab 1 sets up the machine. Lab 7 is the test. Labs 2 to 6 add features.

Grading

Work	Description	Points
Colloquium	End of semester, before the exam session	20
Midterm	Middle of the semester	20
Lab assignments	Labs 1 to 6, cumulative app	30
Practical test	Lab 7, 90 minutes, individual	30
	Pass mark	**50 of 100**

Hint

Each lab is graded when you present it. Be ready to run the app and explain any line.

Installation links

```
# Flutter SDK:
# https://docs.flutter.dev/install
# Add flutter/bin to your PATH

flutter --version
flutter doctor -v

# Android Studio or Xcode
# Pick one IDE: VS Code or IntelliJ
```

Before starting

Install Flutter, an IDE, and either Android Studio or Xcode.

Target

The current stable Flutter channel.

Important

Type URLs exactly. They are complete on one line.

Environment setup

1. Install Flutter from docs.flutter.dev/install.
2. Follow Install manually for your OS.
3. Add flutter/bin to your PATH.
4. Install Android Studio or Xcode (macOS).
5. Install an editor: VS Code or IntelliJ.
6. Run `flutter doctor`, fix every blocking issue.

Done when

- [ ] `flutter --version` works in a new terminal.
- [ ] `flutter doctor` shows no red warnings for Flutter or Android.
- [ ] You can explain one warning you fixed.

Running the default app

1. Boot an emulator or simulator.
2. Run `flutter devices` to confirm it appears.
3. Create the project: `flutter create lab1_app`.
4. Run it: `flutter run`.
5. Change the app title string in `lib/main.dart`.
6. Press r to hot-reload and see it update.

Done when

- [ ] The counter app runs and the button increments it.
- [ ] Your edited title appears after hot reload.
- [ ] You can repeat the hot reload on request.

flutter doctor errors

`cmdline-tools missing`	Android Studio > SDK Manager > SDK Tools > Android SDK Command-line Tools. Install, then run `flutter doctor` again.
`License unknown`	Run `flutter doctor --android-licenses` and answer `y` to all prompts.
`Unable to locate SDK`	Open Android Studio once. The SDK downloads on first launch. On Mac, accept the Xcode license first.

Emulators and devices

`Emulator never boots`	Enable hardware acceleration (VT-x / AMD-V) in BIOS. On Apple silicon Macs, pick arm64 images only.
`No devices listed`	Run `flutter devices`. Start the emulator with `flutter emulators --launch <id>`. Physical Android phones work with USB debugging on.
`Xcode but no simulator`	Open Xcode once and accept the license, then run `open -a Simulator`. If it still fails, run `xcode-select --install`.

Keeping your work in Git

1. Create a repository on GitHub.
2. Clone it: `git clone <url>`.
3. Work on your code, commit often with clear messages.
4. Push your commits: `git push`.

The app grows across seven labs. A commit after each task keeps every working state one command away.

Presenting your work

1. Call the lab instructor when your checklists are done, or in the last twenty minutes.
2. Run the app live, on your emulator or device.
3. Walk through each task and answer questions on your code.

Today

Show `flutter doctor`, the counter app running, and your edited title after a hot reload.

A lab can only be presented in its own session.

Presentation

The app running · one hot reload, shown live